

# The Conative Bridge: Active Inference Meets Integrated Information Theory ( $\Phi$ )

The Ontogenetic Predictive Engine & The 6th Axiom of Consciousness (The Will to Exist)

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## ★ THE FUNDAMENTAL MASTER EQUIVALENCE OF CONSCIOUS AUTOPOIESIS ★

$$\pi^* = \arg \min_{\pi} \sum_{\tau} G(\pi, \tau) \iff \mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\Phi > 0)$$

3rd-Person Physics (Karl Friston): Expected Free Energy Minimization across the Markov Blanket  $\equiv$  1st-Person Phenomenology (Giulio Tononi & Thomas Riebl): Autopoietic Persistence of Causal Power via the 6th Axiom (The Will to Exist)

### ⚡ 1. 3rd-Person Exteriority (FEP)

**Karl Friston (Active Inference & POMDP):** Self-organizing systems preserve physical existence against thermodynamic decay by minimizing **Variational Free Energy ( $F$ )** and selecting policy  $\pi^*$  to minimize **Expected Free Energy ( $G$ )**:

$$F \geq -\ln p(o) \mid G(\pi) = \text{Pragmatic} + \text{Epistemic}$$

- Perceptual Inference:** Updating hidden states  $q(s)$  via prediction error  $\gamma(o - g(s))$ .
- Active Inference:** Executing actions  $u$  to force sensory states  $o$  toward prior homeostatic preferences ( $C$ ).
- Markov Blanket:** Statistical phase-space partition separating Internal States ( $\mu$ ) from External States ( $\eta$ ).

### 🔥 2. The 6th Axiom of IIT (Riebl)

**The Missing Conative Axiom:** Standard IIT 4.0 defines five static axioms (Existence, Intrinsicity, Information, Integration, Exclusion), creating the *Paradox of Transient Causal Phantoms* (static lifeless logic gates having high  $\Phi$ ).

$$\mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\text{Autopoiesis})$$

- Conative Dynamics:** Genuine consciousness requires active temporal self-preservation (Spinoza's *Conatus*, Schopenhauer's *Wille zum Leben*).
- Action-Perception Loop:**  $\Phi$  is sustained because the system actively chooses policies  $\pi^*$  that avoid phase-space collapse ( $\Phi \rightarrow 0$ ).
- The Edge of Chaos:** Metastable balance between differentiation and integration sustaining maximal causal power.

### 🧠 3. 1st-Person Interiority (IIT 4.0)

**Giulio Tononi (Expanded IIT 4.0):** The qualitative interiority of an entity is strictly identical to its **Intrinsic Integrated Causal Structure ( $S$ )** and causal power ( $\Phi > 0$ ) across the Minimum Information Partition (MIP):

$$\Phi(M_1; M_2) = \frac{1}{2} (\ln \det \Sigma_{M_1} + \ln \det \Sigma_{M_2} - \ln \det \Sigma_{\text{Whole}})$$

- Maximal Substrate:** Consciousness exists only at the spatial-temporal grain where integrated cause-effect power  $\Phi$  reaches a local maximum.
- The Dissociated Alter:** A localized whirlpool in Mind-at-Large whose internal  $\Phi$  is decoupled from external states by the Markov Blanket.

### 🧬 Chapter 5 Synthesis: Lifelong Learning & The Predictive Self (The Ontogenetic Tapestry)

How the individual soul develops its unique character, autobiographical narrative, and cognitive depth across ontogenesis:

#### 1. Lived Experience

Sensory stream, relationships, culture & environmental crises

#### 2. Error Minimization

Bayesian updating of Generative Matrices ( $A, B, C$ ) via Active Inference

#### 3. Synaptic Consolidation

Cortical-hippocampal memory engrams & LTP consolidation

#### 4. Narrative Self

Neocortical 'Press Secretary' (Roth) constructing the biographical life story

### 📊 Unified POMDP Parameters & Soul Architecture

|              |                                              |                     |                                    |
|--------------|----------------------------------------------|---------------------|------------------------------------|
| $D_0$ Prior: | Genetics (15%) • Limbic baseline (*Conatus*) | $\gamma$ Precision: | Epigenetics (10%) • Stress gating  |
| $A$ Matrix:  | Lifelong Learning (25%) • Likelihood engrams | $B$ Tensor:         | Biographical World-Simulator (25%) |
| $C$ Vector:  | Homeostatic Preferences • The Will to Exist  | Blanket:            | Dissociative Alter Boundary (10%)  |

\* **Formal Scientific Footnote on the 6th Axiom:** The 6th Axiom ("The Will to Exist / Autopoietic Causal Persistence":  $\mathbb{E}[\Phi(t+1) \mid \pi] \geq \Phi(t)$ ) is **Thomas Riebl's proposed theoretical extension to Integrated Information Theory 4.0 (IIT 4.0 by Tononi et al.)**. While standard IIT 4.0 formulates five static phenomenological axioms (Existence, Intrinsicity, Information, Integration, Exclusion), Riebl's 6th Axiom introduces the necessary *temporal, autopoietic action-perception requirement*. It resolves the "Transient Causal Phantom" paradox and mathematically bridges the static causal topology of IIT with Karl Friston's dynamic Free Energy Principle, establishing that conscious entities are self-sustaining active inference networks driven by an intrinsic Will to Exist.